For categorical variables: χ^2 test

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Categorical data and χ^2 tests

Pearson's χ^2 test

1. Goodness of fit test 拟合优度检验: 分类变量的频率分布是否与我们的预期不同？
2. Test of independence: 两个分类型变量是否相关?

Mathematically, they are actually the same.

χ^2 goodness of fit

For one categorical variable

To test whether the observed distribution of a categorical variable differs from our expectations.

Examples:

We can answer questions like: given the drawn sample, does the population have:

- Equal proportions of male and female?
- 90% right-handed and 10% left-handed people?

χ^2 goodness of fit

Example

Mode of transportation	Observed	Expected
Public	200	250
Drive	300	250

H_0: 50% of citizens prefer to drive, and 50% prefer public transit

H_1: The proportion of citizens who prefer to drive is not 50%, and the proportion preferring public transit is not 50%.

χ^2 goodness of fit

Example

Transportation	Observed	Expected	$O-E$	$(O-E)^2$	$\frac{(O-E)^2}{E}$
Public	200	250	-50	2500	2500/250
Drive	300	250	50	2500	2500/250

We need a measure of goodness of fit, a statistic to capture the difference between the observation and the expectation:

$$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e}$$

χ^2 goodness of fit

The idea

- If the difference is too large, H_0 should be rejected
- If the difference is not large enough, we cannot reject H_0

To find the significance level

$$\sum \frac{(f_o - f_e)^2}{f_e} \sim \chi^2(df)$$

where $df = k - 1$, k is the number of groups

χ^2 goodness of fit

$\chi^2(1)$

- $\alpha = 0.05$ and $df = 1$, the χ^2 critical value ($\chi^2_{0.05}(1)$) is 3.84
- The χ^2 value based on the sample is 20

- **表明实际的交通方式选择与预期的有显著性差异**
- **实际中选择自驾的市民比预期多，选择公交的市民比预期少**

χ^2 test of independence

For two categorical variables

To test whether two categorical variables are related to each other

Example

Gender\Eyesight	Normal	color blindness
Male	140	10
Female	145	5

χ^2 test of independence

The hypothesis

H_0: Gender and eyesight are not related in the population

H_1: Gender and eyesight are related in the population

Or

The hypothesis

H_0: The proportions of color-blinded people are the same for male and female

H_1: The proportions of color-blinded people are not the same for male and female

χ^2 test of independence

Observed values

Gender\Eyesight	Normal	color blindness	Sum
Male	140	10	150
Female	145	5	150
Sum	285	15	300

χ^2 test of independence

If uncorrelated ... ?

Gender\Eyesight	Normal	color blindness	Sum
Male			150
Female			150
Sum	285	15	300

χ^2 test of independence

Expected values

Gender\Eyesight	Normal	Color blindness	Row totals
Male	$142.5(\frac{285}{300} \times 150)$	$7.5(\frac{15}{300} \times 150)$	150
Female	$142.5(\frac{285}{300} \times 150)$	$7.5(\frac{15}{300} \times 150)$	150
Column totals	285	15	$N = 300$

The expected frequency for row r and column c is:

$$\frac{\text{Row } r \text{ total} \times \text{Column } c \text{ total}}{N}$$

χ^2 test of independence

χ^2: Capturing the difference

$$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e}$$

Gender\Eyesight	Normal	Color blindness	Row totals
Male	140(142.5)	10(7.5)	150
Female	145(142.5)	5(7.5)	150
Column totals	285	15	$N = 300$

$$\frac{(140 - 142.5)^2}{142.5} + \frac{(10 - 7.5)^2}{7.5} + \frac{(145 - 142.5)^2}{142.5} + \frac{(5 - 7.5)^2}{7.5}$$
$$= 1.754$$

χ^2 test of independence

χ^2

$$\sum \frac{(f_o - f_e)^2}{f_e} \sim \chi^2(df)$$

where $df =$ (number of variable 1 groups - 1) $\times$ (number of variable 2 groups - 1)

$\chi^2(1)$

- $\alpha = 0.05$ and $df = 1$, the χ^2 critical value is 3.84
- The χ^2 value base on the sample is 1.754

  We fail to reject the null hypothesis

χ^2 test of independence

Degree of freedom

	C1	C2	C3	C4	Row total
R1	✓	✓	✓	*	RT1
R2	✓	✓	✓	*	RT2
R3	*	*	*	*	RT3
Column total	CT1	CT2	CT3	CT4	

$$df = (ncol - 1) \times (nrow - 1)$$

Note: There need to be sufficient observations in each combined group (e.g., a minimum of five).

In practice

Questionnaire： cgss2021

A42.您是否同意以下说法：【出示示卡 3】

	完全不同意	比较不同意	无所谓同意不同意	比较同意	完全同意	不知道	拒绝回答
1. 男人以事业为重，女人以家庭为重	1	2	3	4	5	98	99
2. 男性能力天生比女性强	1	2	3	4	5	98	99
3. 干得好不如嫁得好	1	2	3	4	5	98	99
4. 在经济不景气时，应该先解雇女性员工	1	2	3	4	5	98	99
5. 夫妻应该均等分摊家务	1	2	3	4	5	98	99

Categorical data analysis: "It's better to marry well than to work well."

The original data

	a2 性别	a423 [3 干得好不如嫁得好] 您是否同意以下说法:
1	2	2
2	1	3
3	2	4
4	1	4
5	2	3
6	2	4
7	2	2
8	1	4
9	1	1

Showing 1 to 10 of 12,472 entries, 2 total columns

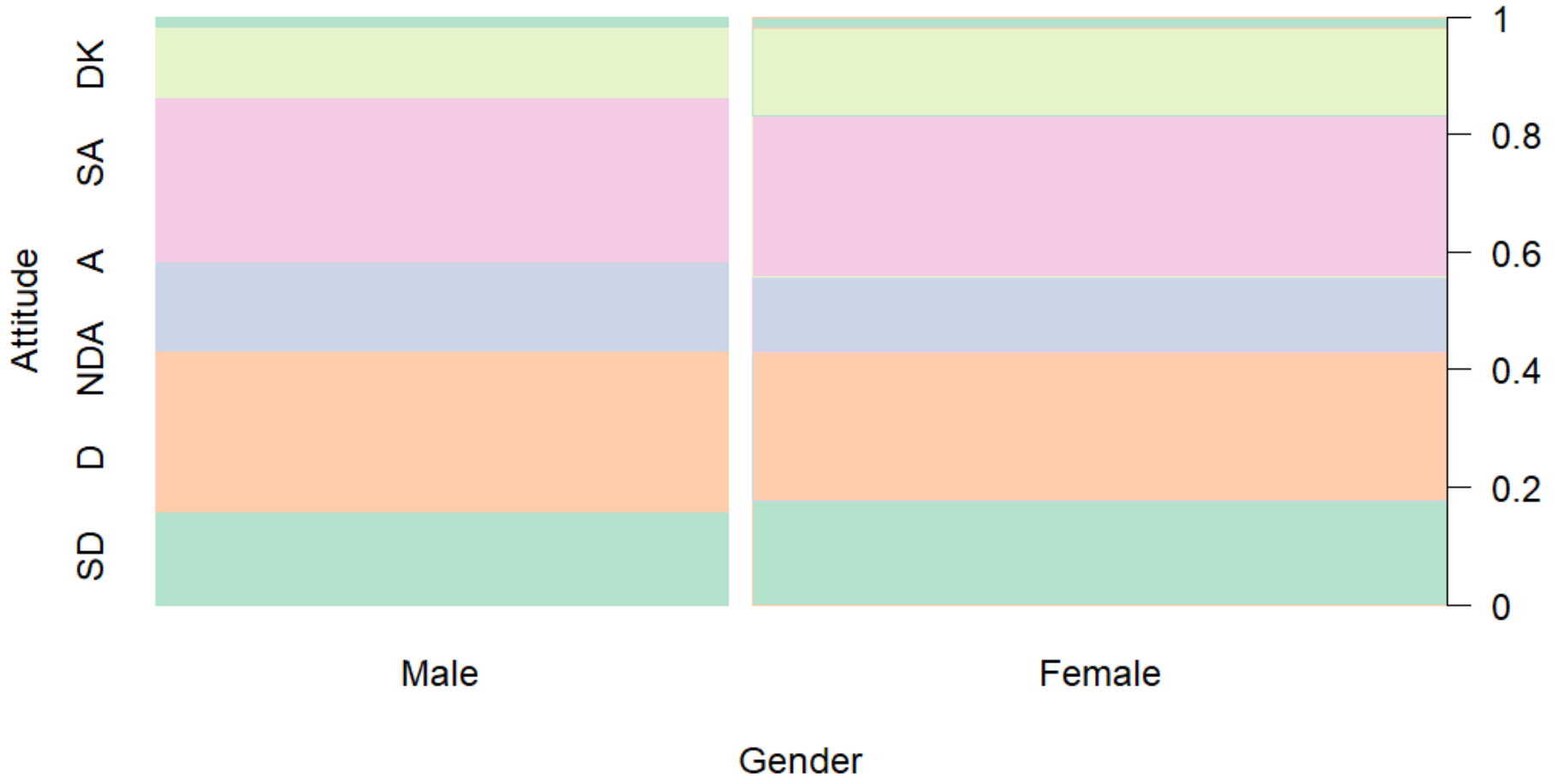
- n=8148
- Gender: 1 Male, 2 Female
- Attitude:
  1 Completely disagree
  2 Disagree
  3 Neither disagree or agree
  4 Agree
  5 Completely agree
  98 Don't know
  99 Refusal

Categorical data analysis: "It's better to marry well than to work well."

Contingency table: observations

	Completely D	Disagree	Neither D or A	Agree	Completely A	DK	Total
Male	585	1008	551	1036	437	62	3679
Female	794	1130	564	1234	664	83	4469
Total	1379	2138	1115	2270	1101	145	8148

Categorical data analysis: "It's better to marry well than to work well."



Contingency table: observations

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Contingency table: expectations

	Completely D	Disagree	Neither D or A	Agree	Completely A	DK	Total
Male	622	965	503	1024	497	65	3679
Female	756	1172	611	1245	603	79	4469
Total	1379	2138	1115	2270	1101	145	8148

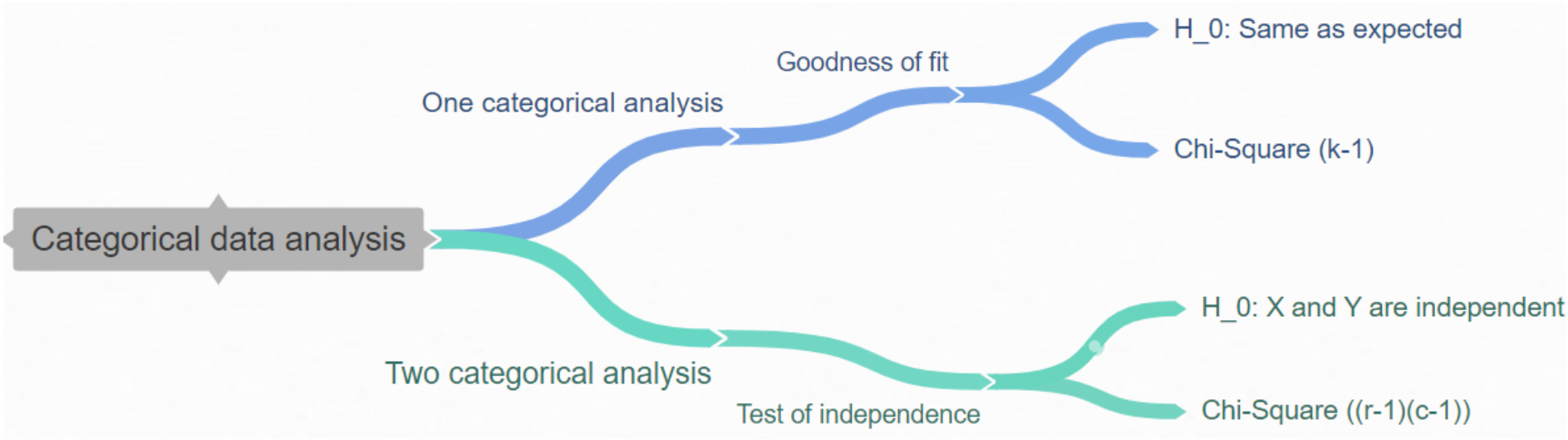
Categorical data analysis: "It's better to marry well than to work well."

χ^2 test

- $\alpha = 0.05$, $df = 5$, and the χ^2 critical value is 11.0705
- The χ^2 based on the survey response is 29.58557

  The null hypothesis is rejected. Gender and gender role attitude are not independent from each other.

Summary


Categorical data analysis
One categorical analysis
Goodness of fit
H_0: Same as expected
Chi-Square (k-1)
Two categorical analysis
Test of independence
H_0: X and Y are independent
Chi-Square ((r-1)(c-1))